

# An intercity Sr/Yb optical clock comparison at the $2.1 \times 10^{-18}$ uncertainty over a 1350 km fiber link

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## Abstract

Redefining the International System of Units (SI) second requires independent frequency-ratio measurements between different types of optical clocks in different laboratories, with an overall measurement uncertainty below  $5 \times 10^{-18}$  and consistent results. However, although several optical clocks have reported self-evaluated uncertainties below  $2 \times 10^{-18}$ , validation of these self-assessed values through frequency-ratio comparisons has not yet been achieved. In this study, we compared a ytterbium optical lattice clock in Wuhan with a strontium optical lattice clock in Shanghai—approximately 700 km away—using a phase-stabilized fiber link spanning 1350 km. Based on 274 hours of data collected over 58 consecutive days, we measured the Yb/Sr frequency ratio as  $\text{Yb/Sr} = 1.207\,507\,039\,343\,337\,720\,6(21)$ , with a Bayesian statistical uncertainty of  $0.89 \times 10^{-18}$  and a total uncertainty of  $2.1 \times 10^{-18}$ . The value differs from that of NIST–JILA  $1.207\,507\,039\,343\,337\,723\,0(37)$  by  $-2.44 \times 10^{-18}$ , consistent within the margin of statistical error. To our knowledge, this is the first intercity, cross-species frequency-ratio measurement to meet the uncertainty requirements outlined in the roadmap for redefining the SI second. Furthermore, together with the NIST–JILA frequency-ratio result, this achievement satisfies the mandatory criteria (I.2.b) for the redefinition of the second, confirming the validity of uncertainty assessments for both ytterbium and strontium atomic clocks.

## 1 Introduction

Optical lattice and single-ion clocks now reach fractional systematic uncertainties at or below  $2 \times 10^{-18}$ —including ytterbium<sup>1;2</sup> and strontium<sup>3–5</sup> lattice clocks and  $\text{Al}^{+6;7}$ ,  $\text{Lu}^{+8}$  and  $\text{Ca}^{+9}$  ion clocks—two orders of magnitude better than current primary frequency standards<sup>10;11</sup>. By comparing independent optical clocks located in different regions and laboratories, the precision budget self-assessed by individual clocks can be verified more objectively. This is because the environmental conditions and geodetic assumptions used in self-evaluation are typically based on a single location, and using identical methods within a single laboratory may lead to common systematic biases. Therefore, connecting clocks from different locations and types into a collaborative network is a key pathway toward establishing optical clocks as a verifiable metrological infrastructure<sup>12;13</sup>. The idea has moved from proposal to practice: fiber links with noise levels below that of the clocks themselves now span thousands of kilometers<sup>14;15</sup>, and multi-clock campaigns have measured tens of ratios among clocks in six countries<sup>16</sup> and across a continental network<sup>17</sup>. Satellite-based advanced links with comparable performance are also being developed and will interconnect with fiber networks to form a future global optical clock network<sup>18–20</sup>.

The most immediate purpose of such a network is the redefinition of the SI second. One of the mandatory requirements (I.2.b) in the roadmap for this redefinition<sup>21</sup> is that frequency ratios of different clock types be measured at least twice by different institutes in agreement with an overall uncertainty of the comparison  $\Delta\nu/\nu < 5 \times 10^{-18}$ —a requirement reiterated in the 2025 CCTF progress report<sup>22;23</sup>. However, comparisons across independent institutions and different locations have reached at best a precision of  $7.7 \times 10^{-18}$ <sup>17</sup>. So far, no frequency-ratio measurements between clocks in different cities have satisfied this requirement<sup>24</sup>.



Figure 1: The two-clock network. An IAPMST ytterbium clock in Wuhan and a USTC strontium clock in Shanghai are connected by a phase-stabilized 1550 nm fiber link. Relay stations along the route carry erbium-doped fiber amplifiers and fiber-noise-cancellation loops.

Closing that gap matters beyond the definition itself. Chronometrically leveled clocks map the gravitational potential of the Earth at centimeter resolution<sup>1;25;26</sup>; clock networks have been proposed and used as sensors for dark matter<sup>27;28</sup>. All of these applications assume that remote clocks agree at a level that, until now, has only been demonstrated locally twice: a Sr/Yb/Al<sup>+</sup> triple-clock ratio measurement with an uncertainty of  $3.2 \times 10^{-18}$  by NIST–JILA in Boulder (US)<sup>29</sup>, and a Yb<sup>+</sup>/Sr measurement with  $4.3 \times 10^{-18}$  by PTB in Braunschweig (Germany)<sup>30</sup>.

Here we report an intercity, cross-species frequency-ratio measurement between an IAPMST ytterbium lattice clock in Wuhan and a USTC strontium lattice clock in Shanghai, about 700 km apart, connected by a 1 350 km fiber link. We determine the Yb/Sr ratio with a total uncertainty of  $2.1 \times 10^{-18}$ ; gravitational redshift effects have been corrected, whereas measurement error and the time-varying gravitational redshift term associated with gravitational tides have been retained as uncertainty components. The result establishes the two sites as a prototype node pair for a distributed optical-clock network, and completes, at intercity scale, the class of ratio measurements that the redefinition process depends on.

## 2 Results

### 2.1 Network and observation campaign

Figure 1 shows the two-clock network. The Yb and Sr clocks are separated by 700 km and connected through a 1 350 km phase-stabilized fiber link operated at 1550 nm; the loop-back topology and the relay stations keep the link’s added instability below the clock noise for averaging times beyond 100 s. Over two months (58 days, 3 stages) the two clocks and their transfer chains ran synchronously for a total of 1 008 912 valid seconds, recorded as 17 jump-free segments; the data set analyzed here comprises 16 segments with 986 416 seconds (274 effective hours). Figure 2a places the segments on the campaign timeline.

### 2.2 Clock-ratio determination

We determine the Yb/Sr frequency ratio from the 16 segments. Each segment yields a ratio estimate with its own statistical uncertainty estimated from the Allan deviation of the beat (Methods). The segment estimates scatter more than their individual uncertainties allow: the fixed-effect weighted least-squares fit gives  $\chi^2_{\text{red}} = 4.017$  for 15 degrees of freedom ( $p = 2.3 \times 10^{-7}$ ), and 5 of the 16 segments have stability fits that fall outside the white-frequency-noise band (groups 2, 6, 7, 12, 16). We therefore adopt the Bayesian random-effect estimate, which treats

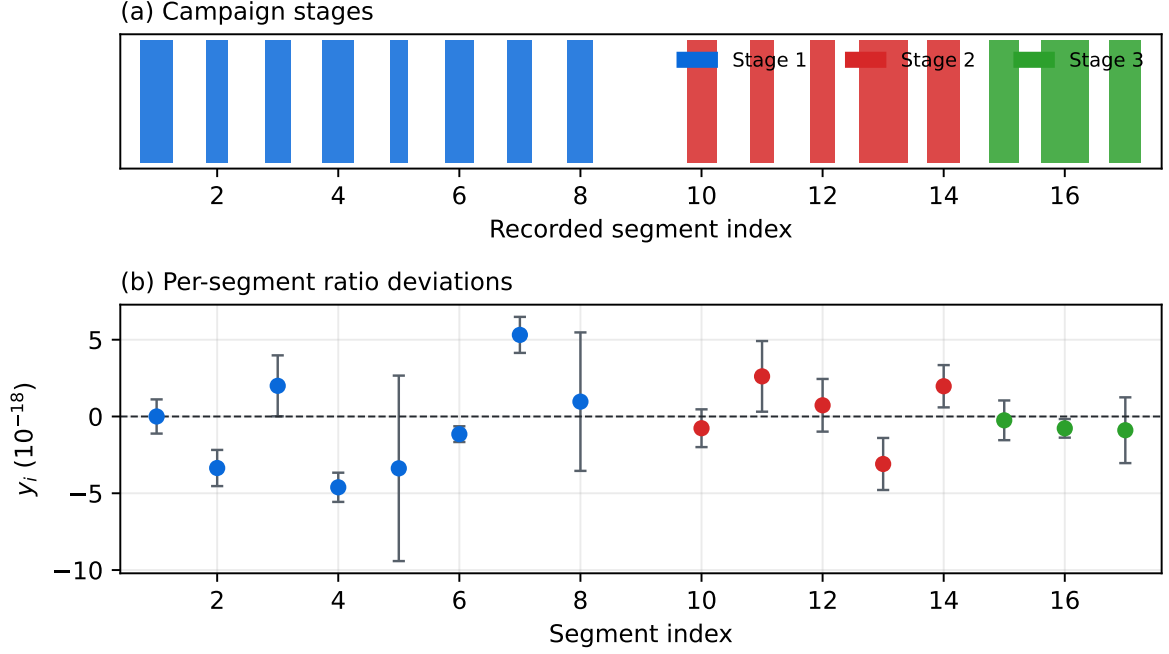


Figure 2: Campaign and per-segment data. (a) Recorded segments colored by measurement stage; bar widths scale with the square root of the retained sample count. (b) Per-segment ratio deviations  $y_i$  with their statistical uncertainties; the dashed line marks zero deviation.

the excess scatter as a free parameter and propagates its posterior uncertainty, while the fixed-effect WLS and Mandel–Paule estimates are kept as cross-checks.

The resulting Yb/Sr ratio is

$$\text{Yb/Sr} = 1.207\,507\,039\,343\,337\,720\,6(21), \quad (1)$$

with a statistical uncertainty of  $0.89 \times 10^{-18}$  and a total standard uncertainty of  $2.1 \times 10^{-18}$  (Section 2.3). The precision-weighted cross-checks agree within their uncertainties: the WLS center is 1.207 507 039 343 337 720 0 and the Mandel–Paule center is 1.207 507 039 343 337 720 6 (rounded at the 19th decimal), both consistent with the adopted value at the  $10^{-18}$  level. Figure 2b shows the per-segment deviations; the segment-to-segment scatter is dominated by the statistical noise of the individual segments and by the excess scatter captured by the random-effect term.

### 2.3 Uncertainty budget

Table 1 collects the components of the total standard uncertainty (Figure 3b). The statistical term dominates the budget, followed by the Yb and Sr systematic evaluations, the static-potential term from the leveling, and the time-varying-redshift term carried as an uncertainty item; the fiber-link and comb terms are negligible at this precision. The total standard uncertainty is  $2.1 \times 10^{-18}$ .

## 3 Discussion and outlook

The value reported here agrees with the two existing measurements of the Yb/Sr ratio at the level set by the redefinition roadmap (Figure 3a). Relative to the NIST–JILA value  $1.207\,507\,039\,343\,337\,723\,0(37)^{29}$ , our central value differs by  $-2.44 \times 10^{-18}$ ; relative to the European-network value  $1.207\,507\,039\,343\,337\,718(32)^{17}$ , it differs by  $+2.56 \times 10^{-18}$ . Both differences lie within the  $5 \times 10^{-18}$  agreement threshold of the

Table 1: Uncertainty budget of the Yb/Sr frequency-ratio measurement. The time-varying gravitational-redshift term is carried as an uncertainty item and is not applied as a correction to the central value. Components are treated as independent.

| Component                            | Contribution ( $10^{-18}$ ) |
|--------------------------------------|-----------------------------|
| Statistical (Bayesian random effect) | 0.885                       |
| Sr systematics                       | 0.92                        |
| Yb systematics                       | 1.3                         |
| Static gravitational potential       | 0.946                       |
| Fiber link                           | 0.1                         |
| Frequency comb                       | 0.1                         |
| Time-varying redshift                | 0.5                         |
| Total (quadrature)                   | 2.12                        |

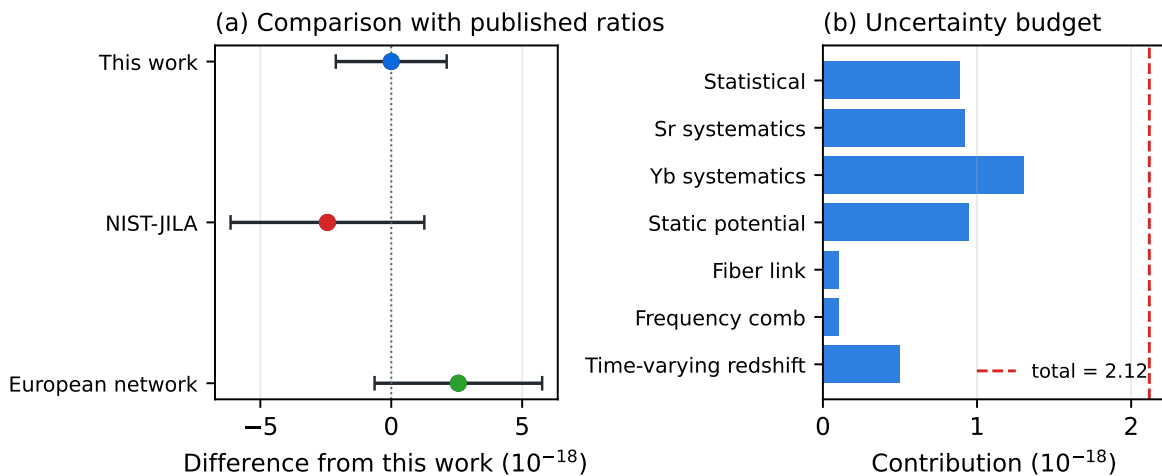


Figure 3: Comparison and budget. (a) Differences between this work and the published Yb/Sr ratios, with their quoted uncertainties. (b) Components of the total standard uncertainty; the dashed line marks the quadrature total.

roadmap’s frequency-ratio criterion<sup>21;23</sup>. To our knowledge, this is the first reported intercity (100-km-scale), cross-species frequency-ratio measurement meeting that uncertainty requirement (as of *October 2026*); existing ratios at or below the threshold were measured on a single campus<sup>29</sup>, while inter-institute comparisons through fiber networks have so far reached  $7.7 \times 10^{-18}$ <sup>17</sup>. The result closes, at intercity scale, the class of frequency-ratio measurements that the SI-second redefinition process depends on. Meeting the roadmap’s frequency-ratio requirement has so far required clocks on a shared campus, where a single stabilized link serves both systems; here the two clocks sit in different cities, in different institutions, and use different atomic species, and the measurement still reaches the threshold, because the fiber link and the two combs contribute well below the clocks themselves. The bottleneck for reaching the redefinition threshold is therefore no longer the transfer infrastructure but the systematic evaluations of the clocks and the geodetic determination of their potential difference.

Changes in the Earth’s gravitational potential enter the comparison directly. The static difference between the two clock transitions is fixed by first-order spirit leveling and is carried in the uncertainty budget; of the time-varying contributions, the tidal term is carried as a data-driven uncertainty item, while co-seismic steps from the regional seismicity during the campaign fall well below the reported precision. At this accuracy, it is the geodetic determination of the

potential difference, rather than its variability, that limits the comparison; the same sensitivity underlies proposed chronometric-geodesy applications of optical-clock networks<sup>1;25;26</sup>.

The campaign also demonstrates the operational character that a future optical-clock network will need. Over the two-month campaign the two clocks were restarted, the transfer chain re-locked, and the beat record interrupted many times; the segment-level ratio estimates show no evidence of a systematic offset between the three measuring stages, so the reproducibility of the comparison does not depend on continuous operation of either system. This is the practical counterpart of the coordinated multi-clock campaigns<sup>16;17</sup>: the network tolerates interruption, provided each retained window is self-consistently characterized.

Several limitations bound the present result. First, the per-segment statistical uncertainties from the Allan-deviation extrapolation are the dominant budget term, and five of the sixteen segments carry model-limited fits; the random-effect treatment accounts for their excess scatter at the level of the reported uncertainty, but a longer record would reduce the statistical term. Second, the systematic evaluations of the two clocks are those published by the laboratories; independent verification of those budgets, as opposed to the transfer, is not attempted here. Third, the geopotential difference is taken from a first-order leveling campaign; an alternative value exists in the experiment records and is carried as an external item pending reconciliation, as is the physical-model provenance of the supplied tide series and the exact accounting of the full sample totals. Finally, the present campaign meets the uncertainty requirement of the roadmap but not the full criterion, which additionally requires each ratio to be measured at least twice by different institutes and repeated over an extended period.

Looking ahead, the result establishes this link as a prototype node pair of a distributed optical-clock network: the same infrastructure can be extended to further nodes, and the technique demonstrated here – clocks at different sites agreeing at the redefinition threshold – is the operation that both the redefinition and post-redefinition applications require. Realizing those applications over larger distances will mean combining fiber networks of the kind used here<sup>14;15</sup> with intercontinental techniques such as VLBI<sup>31</sup>, and will put the same demands on geodetic infrastructure that this comparison faced<sup>1;25</sup>.

## References

- [1] W. F. McGrew, X. Zhang, R. J. Fasano, S. A. Schäffer, K. Beloy, D. Nicolodi, R. C. Brown, N. Hinkley, G. Milani, M. Schioppo, T. H. Yoon, and A. D. Ludlow. Atomic clock performance enabling geodesy below the centimetre level. *Nature*, 564:87–90, 2018. doi: 10.1038/s41586-018-0738-2. arXiv:1807.11282.
- [2] Qiang Zhu, Jingran Shi, Yuechen Zhang, Xiatian Xu, Bing Wang, Dezhi Xiong, Zhuanxian Xiong, Pengcheng Fang, Qunfeng Chen, Lingxiang He, and Baolong Lyu. Ytterbium lattice clock with systematic uncertainty of  $1.3 \times 10^{-18}$  and instability at the  $10^{-19}$  level. *arXiv preprint*, 2026. arXiv:2606.10514; submitted to Metrologia.
- [3] Alexander Aeppli, Kyungtae Kim, William Warfield, Marianna S. Safronova, and Jun Ye. Clock with  $8 \times 10^{-19}$  systematic uncertainty. *Phys. Rev. Lett.*, 133:023401, 2024. doi: 10.1103/PhysRevLett.133.023401.
- [4] Zhi-Peng Jia, Jie Li, De-Quan Kong, Xiang Zhang, Hai-Wei Yu, Xiao-Yong Liu, Yu-Chen Zhang, Yuan-Bo Wang, Xian-Qing Zhu, Jia-Hao Zhang, Ming-Yi Zhu, Pei-Jun Feng, Xing-Yang Cui, Ping Xu, Xiao Jiang, Xiang-Pei Liu, Peng Liu, Han-Ning Dai, Yu-Ao Chen, and Jian-Wei Pan. Improved systematic evaluation of a strontium optical clock with uncertainty below  $1 \times 10^{-18}$ . *Metrologia*, 63:025002, 2026. doi: 10.1088/1681-7575/ae449e.
- [5] Xiao-Tong Lu, F. Guo, Y.-Y. Liu, J. Cao, J.-A. Li, J.-J. Xia, Q.-F. Xu, B.-Q. Lu, Y.-

- B. Wang, and H. Chang. NTSC SrII optical lattice clock with uncertainty of  $2 \times 10^{-18}$ . *Metrologia*, 62(3):035007, 2025. doi: 10.1088/1681-7575/addc77.
- [6] Mason C. Marshall, Daniel A. Rodriguez Castillo, Willa J. Arthur-Dworschack, Alexander Aepli, Kyungtae Kim, Dahyeon Lee, William Warfield, Joost Hinrichs, Nicholas V. Nardelli, Tara M. Fortier, Jun Ye, David R. Leibbrandt, and David B. Hume. High-stability single-ion clock with  $5.5 \times 10^{-19}$  systematic uncertainty. *Phys. Rev. Lett.*, 135:033201, 2025. doi: 10.1103/hb3c-dk28.
  - [7] Fabian Dawel, Johannes Kramer, Derwell Drapier, Lennart Pelzer, Kai Dietze, Mirza A. Ali, Marek Hild, Vincent Barbé, Steven A. King, Joshua Klose, Kilian Stahl, Johannes Rahm, Navraj Poudel, Sören Dörscher, Stefan Weyers, Erik Benkler, Christian Lisdat, and Piet O. Schmidt. An  $\text{Al}^+$  clock with  $1.6 \times 10^{-18}$  systematic uncertainty and its frequency ratios, 2026. URL <https://arxiv.org/abs/2606.23151>.
  - [8] K. J. Arnold, M. D. K. Lee, Qi Zhao, Qichen Qin, Zhao Zhang, N. Jayjong, and M. D. Barrett.  $\text{Lu}^+$  optical frequency references with accuracy verified at the 19th digit. *Nature*, 2026. doi: 10.1038/s41586-026-11072-8.
  - [9] Bao-lin Zhang, Zi-xiao Ma, Yao Huang, Hui-li Han, Ru-ming Hu, Yu-zhuo Wang, Hua-qing Zhang, Li-yan Tang, Ting-yun Shi, Hua Guan, and Ke-lin Gao. Liquid-nitrogen-cooled  $^{40}\text{Ca}^+$  ion optical clock with a systematic uncertainty of  $4.4 \times 10^{-19}$ . *Phys. Rev. Lett.*, 136:053202, 2026. doi: 10.1103/vngc-c1xv.
  - [10] Thomas P. Heavner, Elizabeth A. Donley, Filippo Levi, Giovanni Costanzo, Thomas E. Parker, Jon H. Shirley, Neil Ashby, Stephan Barlow, and Steven R. Jefferts. First accuracy evaluation of NIST-F2. *Metrologia*, 51(3):174–182, 2014. doi: 10.1088/0026-1394/51/3/174.
  - [11] S. Weyers, V. Gerginov, M. Kazda, J. Rahm, B. Lipphardt, G. Dobrev, and K. Gibble. Advances in the accuracy, stability, and reliability of the PTB primary fountain clocks. *Metrologia*, 55(6):789–805, 2018. doi: 10.1088/1681-7575/aae008.
  - [12] Fritz Riehle. Optical clock networks. *Nature Photonics*, 11(1):25–31, 2017. doi: 10.1038/nphoton.2016.235.
  - [13] P. Kómár, E. M. Kessler, M. Bishof, L. Jiang, A. S. Sørensen, J. Ye, and M. D. Lukin. A quantum network of clocks. *Nature Physics*, 10(8):582–587, 2014. doi: 10.1038/nphys3000.
  - [14] Marco Schioppo, Jochen Kronjäger, Alissa Silva, Rada Ilieva, Chris W. Oates, Jacopo Tunesi, Charles F. A. Baynham, Helen S. Margolis, and Rachel M. Godun. Comparing ultrastable lasers at  $7 \times 10^{-17}$  fractional frequency instability through 2220 km of optical fiber. *Nature Communications*, 13:212, 2022. doi: 10.1038/s41467-021-27884-3.
  - [15] Fa-Xi Chen, Li-Bo Li, Jiu-Peng Chen, Kan Zhao, Jian-Yu Guan, Yang Xu, Lei Hou, Fei Zhou, Cheng-Zhi Peng, Qiang Zhang, Hai-Feng Jiang, and Jian-Wei Pan.  $10^{-21}$ -level optical frequency dissemination over 2067 km of noise-loaded field-deployed fiber network. *Light: Science & Applications*, 15:276, 2026. doi: 10.1038/s41377-026-02299-1.
  - [16] T. Lindvall, M. Pizzocaro, R. M. Godun, M. Abgrall, others, and H. S. Margolis. Coordinated international comparisons between optical clocks connected via fiber and satellite links. *Optica*, 12(6):843, 2025. doi: 10.1364/optica.561754.
  - [17] Marco Pizzocaro, Clara Zyskind, Anne Amy-Klein, Erik Benkler, Sebastien Bize, Davide Calonico, Etienne Cantin, Christian Chardonnet, Cecilia Clivati, Stefano Condio, E. Anne Curtis, Simone Donadello, Sören Dörscher, Chen-Hao Feng, Melina Filzinger, Jacques-Olivier Gaudron, Rachel M. Godun, Irene Goti, Ian R. Hill, Wei Huang, Nils Huntemann,

- Matthew Johnson, Joshua Klose, Jochen Kronjäger, Alexander Kuhl, Rodolphe Le Targat, Filippo Levi, Burghard Lipphardt, Christian Lisdat, Jerome Lodewyck, Olivier Lopez, Helen S. Margolis, Maxime Mazouth-Lauro, Alberto Mura, Benjamin Pointard, Paul-Eric Pottie, Matias Risaro, Billy I. Robertson, Marco Schioppo, Kilian Stahl, Martin Steinell, Alexandra Tofful, Mads Tø nnes, and Jacob Tunes. International optical clock comparison using the european optical fiber network. *Phys. Rev. Research*, 8:033250, 2026. doi: 10.1103/14bh-ryxs. arXiv:2604.27963.
- [18] Qi Shen, Jian-Yu Guan, Ting Zeng, Qi-Ming Lu, Liang Huang, Yuan Cao, Jiu-Peng Chen, Tian-Qi Tao, Jin-Cai Wu, Lei Hou, Sheng-Kai Liao, Ji-Gang Ren, Juan Yin, Jian-Jun Jia, Hai-Feng Jiang, Cheng-Zhi Peng, Qiang Zhang, and Jian-Wei Pan. Experimental simulation of time and frequency transfer via an optical satellite-ground link at  $10^{-18}$  instability. *Optica*, 8(4):471–476, 2021. doi: 10.1364/OPTICA.413114.
- [19] Emily D. Caldwell, Jean-Daniel Deschenes, Jennifer Ellis, William C. Swann, Benjamin K. Stuhl, Hugo Bergeron, Nathan R. Newbury, and Laura C. Sinclair. Quantum-limited optical time transfer for future geosynchronous links. *Nature*, 618:721–726, 2023. doi: 10.1038/s41586-023-06032-5.
- [20] Qi Shen, Jian-Yu Guan, Ji-Gang Ren, Ting Zeng, Lei Hou, Meng Li, Yuan Cao, Jiu-Jiang Han, Meng-Yang Lian, Yin Chen, Xuan-Xuan Peng, Chao-Ze Wang, Deng-Ke Zou, Zhao Wang, Yu Wang, Gan-Sheng Pan, Juan Yin, Qiang Zhang, Cheng-Zhi Peng, and Jian-Wei Pan. Free-space dissemination of time and frequency with  $10^{-19}$  instability over 113 km. *Nature*, 610:661–666, 2022. doi: 10.1038/s41586-022-05228-5.
- [21] N. Dimarcq, M. Gertsvolf, G. Mileti, S. Bize, C. W. Oates, E. Peik, D. Calonico, T. Ido, P. Tavella, et al. Roadmap towards the redefinition of the second. *Metrologia*, 61(1):012001, 2024. doi: 10.1088/1681-7575/ad17d2. arXiv:2307.14141.
- [22] Comité Consultatif du Temps et des Fréquences (CCTF). CCTF strategy 2025–2035. Bureau International des Poids et Mesures, 2025. URL <https://www.bipm.org/documents/20126/2071143/CCTF+Strategy.pdf>.
- [23] Marina Gertsvolf. Task force on the redefinition of the second: Progress report. Comité Consultatif du Temps et des Fréquences (CCTF), BIPM, 2025. URL [https://www.bipm.org/documents/20126/273113451/1\\_CCTF\\_TE\\_28042025\\_Gertsvolf/667adba6-fd36-be37-5237-9be9ccc24d06](https://www.bipm.org/documents/20126/273113451/1_CCTF_TE_28042025_Gertsvolf/667adba6-fd36-be37-5237-9be9ccc24d06).
- [24] Tara M. Fortier, Andre N. Luiten, and Helen S. Margolis. Optical atomic clocks: defining the future of time and frequency metrology. *Advances in Optics and Photonics*, 2026. in press.
- [25] Jacopo Grotti, Silvio Koller, Stefan Vogt, Sebastian Häfner, Uwe Sterr, Christian Lisdat, Heiner Denker, Christian Voigt, Ludger Timmen, Antoine Rolland, Fred N. Baynes, Helen S. Margolis, Michał Zawada, Ursula Kleinevoss, Nils Huntemann, Ekkehard Peik, Marco Pizzocaro, Simone Donadello, Guglielmo M. Tino, Filippo Levi, Davide Calonico, Alberto Iaco, Marco Tarallo, Elio Rocco, Thomas König, Ali Al-Masoudi, Tanja Mehlstäubler, Christian Lück, Britta Borchert, and Harald Schnatz. Geodesy and metrology with a transportable optical clock. *Nature Physics*, 14:437–441, 2018. doi: 10.1038/s41567-017-0042-3.
- [26] Masao Takamoto, Ichiro Ushijima, Noriaki Ohmae, Toshihiro Yahagi, Kensuke Kokado, Hisa-aki Shinkai, and Hidetoshi Katori. Test of general relativity by a pair of transportable optical lattice clocks. *Nature Photonics*, 14:411–415, 2020. doi: 10.1038/s41566-020-0619-8.
- [27] Andrei Derevianko and Maxim Pospelov. Hunting for topological dark matter with atomic clocks. *Nature Physics*, 10(12):933–936, 2014. doi: 10.1038/nphys3137.

- [28] P. Wcisło, P. Ablewski, K. Beloy, S. Bilicki, M. Bober, R. Brown, R. Fasano, R. Ciurylo, H. Hachisu, K. Hosaka, et al. New bounds on dark matter coupling from a global network of optical atomic clocks. *Science Advances*, 4(12):eaau4869, 2018. doi: 10.1126/sciadv.aau4869.
- [29] Alexander Aepli, Willa J. Arthur-Dworschack, Kyle Beloy, Caitlin M. Berry, Tobias Bothwell, Angela Folz, Tara M. Fortier, Tanner Grogan, Youssef S. Hassan, Zoey Z. Hu, David B. Hume, Benjamin D. Hunt, Kyungtae Kim, Amanda Koepke, Dahyeon Lee, David R. Leibbrandt, Ben Lewis, Andrew D. Ludlow, Mason C. Marshall, Nicholas V. Nardelli, Harikesh Ranganath, Daniel A. Rodriguez Castillo, Jeffrey A. Sherman, Jacob L. Siegel, Suzanne Thornton, William Warfield, and Jun Ye. Atomic clock frequency ratios with fractional uncertainty  $\leq 3.2 \times 10^{-18}$ . *Phys. Rev. Lett.*, 137:033201, 2026. doi: 10.1103/g865-9mk1. arXiv:2512.21428.
- [30] Chetan Vishwakarma, Ingo Nosske, Tim Lücke, Martin Steinle, Melina Filzinger, Erik Benkler, Sören Dörscher, Nils Huntemann, and Christian Lisdat. Interspecies clock comparison below  $5 \times 10^{-18}$  uncertainty with a transportable clock, 2026. URL <https://arxiv.org/abs/2608.01916>.
- [31] Marco Pizzocaro, Mamoru Sekido, Kazuhiro Takefuji, Hideki Ujihara, Hidekazu Hachisu, Nils Nemitz, Masanori Tsutsumi, Tetsuro Kondo, Eiji Kawai, Ryuichi Ichikawa, Kunitaka Namba, Yoshihiro Okamoto, Rumi Takahashi, Junichi Komuro, Cecilia Clivati, Filippo Bregolin, Piero Barbieri, Alberto Mura, Elena Cantoni, Giancarlo Cerretto, Filippo Levi, Giuseppe Maccaferri, Mauro Roma, Claudio Bortolotti, Monia Negusini, Roberto Ricci, Giampaolo Zacchiroli, Juri Roda, Julia Leute, Gérard Petit, Federico Perini, Davide Calonico, and Tetsuya Ido. Intercontinental comparison of optical atomic clocks through very long baseline interferometry. *Nature Physics*, 17:223–227, 2021. doi: 10.1038/s41567-020-01038-6.

## Methods

### Experimental setup and frequency transfer

The comparison links two optical lattice clocks at different institutions: an ytterbium (Yb) clock at IAPMST in Wuhan and a strontium (Sr) clock at USTC in Shanghai, about 700 km apart. The two sites are connected by a phase-stabilized fiber link of 1 350 km single-pass length (2 700 km loop-back), operated at 1550 nm. The route is divided into spans of tens of kilometers, each terminated by a relay station that carries a narrow-linewidth laser together with an erbium-doped fiber amplifier and a fiber-noise-cancellation loop; the added instability of the link is kept below the clock noise level for averaging times beyond 100 s.

Both ends use independent optical frequency combs to bridge the spectral gap between the clock transitions and the telecom carrier. On the Shanghai side the Sr 698 nm transition is transferred via a 1397 nm continuous laser to the 1550 nm carrier; on the Wuhan side the received carrier is linked to the Yb 578 nm transition through the 1156 nm light. All local frequency references and phase-locked loops are derived from the comb repetition rate  $f_{\text{rep}} = 200$  MHz. The observable is the out-of-loop beat between the received carrier and the local Yb-referenced comb, converted to fractional units with the transfer frequency  $F_{1550} = 193.3992$  THz (see Appendix A).

### Observation campaign and per-segment data

The campaign spanned 58 days, from 29 June to 26 August 2026, in 3 measurement stages. Synchronous-operation windows were first defined from the lock status of the two clocks and of the beat signals; within each window the longest continuous valid record was retained. The

selection used lock status only and did not refer to the resulting ratio deviations. In total 17 jump-free segments were recorded, carrying 1 008 912 valid seconds; the data set analyzed here comprises 16 segments with 986 416 valid seconds (274 effective hours).

## Clock-ratio determination and statistical model

Each retained segment yields a clock-ratio estimate  $R_i$  from the mean of its retained beat samples, inverted through the fixed frequency-chain relation (Appendix A) and corrected for the static gravitational redshift  $\delta_g = -3.116 \times 10^{-15}$  derived from the leveled geopotential difference (see below). The per-segment statistical uncertainty  $u_i$  is obtained from the overlapping Allan deviation of the beat fractional frequency: a  $\sigma_y \propto \tau^{-1/2}$  fit over the range  $128 \text{ s} \leq \tau \leq 0.25 T_i$ , extrapolated to the segment duration  $T_i$ .

The per-segment uncertainties do not fully explain the observed scatter between segments, so the ratio is combined with four estimators: fixed-effect weighted least squares (WLS, weights  $1/u_i^2$ ), Birge scaling, the Mandel–Paule random-effect model, and a Bayesian random-effect model in which the excess scatter  $\xi$  is a free parameter whose posterior uncertainty is propagated into the final result. The Bayesian random-effect center is adopted as the primary statistic; the others are reported as cross-checks. Five of the 16 segments (groups 2, 6, 7, 12, 16) have stability fits that fall outside the white-frequency-noise band and are carried with a model-limited flag; this is one motivation for the random-effect treatment. The full model definitions are given in Appendix C.

## Leveling determination of the geopotential difference

The static gravitational-redshift correction is fixed by the geopotential difference between the two clocks’ atomic transition positions, determined by first-order spirit leveling together with simultaneous gravimetry. A leveling line of about 1 113 km was run between the Wuhan benchmark WS-03 (at IAPMST) and the Shanghai benchmark WS-01 (at USTC) through Ezhou, Huangshi, Huanggang, Lu’an, Hefei, Wuhu, Guangde and Suzhou, using digital levels with sight lengths below 30 m and gravity observed at every leveling point with two relative gravimeters. The leveled normal heights were converted to orthometric heights for the geopotential integration,  $\Delta W_{AB} = -\sum_i g_i \Delta H_i$ , where  $g_i$  is the mean gravity and  $\Delta H_i$  the orthometric height difference of the  $i$ -th segment. The resulting geopotential difference between the two atomic transitions is  $\Delta W = 280.042 \pm 0.085 \text{ m}^2 \text{ s}^{-2}$ . The per-kilometer misclosure of the leveling is 0.255 mm, below the first-order limit.

## Uncertainty budget

Table 1 lists the uncertainty components of the Yb/Sr ratio. The statistical term is the posterior standard deviation of the Bayesian random-effect model applied to the 16-segment data. The Sr and Yb systematic terms are the published evaluations of the two clocks<sup>2,4</sup>; the value adopted for the Yb clock resolves an internal-versus-published discrepancy in favor of the published evaluation. The static-potential term follows from the leveling uncertainty. The fiber-link and frequency-comb terms are experiment-side upper bounds. The time-varying gravitational-redshift term is carried as an uncertainty item, quantified from the scale over which the segment-averaged shift varies across the campaign, and is not applied as a correction to the reported central value. The components are treated as independent and combined in quadrature.

## Data availability

The frozen, machine-readable result layer underlying this manuscript is archived in the repository (<https://github.com/hjiang555-a11y/gravitation-whsh>) under `results/audit-v1/verified/`:

the manifest with SHA-256 15725f5246e87af5a450c8e31ccf0d0ef277d517a493ed1fb44d65a4b86564d4, the derived ledgers and statistics, and the generated audit report. The raw beat files and the leveled geopotential inputs are not redistributed.

## Code availability

The analysis and manuscript pipelines are in the same repository under the Apache License 2.0. The manuscript numbers are generated by `paper/rev2/tools/build_numbers.py`, which reads only the frozen manifest and the curated constants listed in `generated/numbers_manifest.json`.

## A Frequency chain and ratio inversion

The observable is the out-of-loop beat between the transferred 1550 nm carrier and the local Yb-referenced comb. All frequencies in the chain derive from the comb repetition rate  $f_{\text{rep}} = 200$  MHz. The Sr clock’s absolute frequency  $f_{\text{Sr}}$  (698 nm) is related to the comb frequency fed to the 1397 nm branch by

$$f_{\text{comb}}^{1397} = \frac{1}{2} (1 + a) f_{\text{Sr}}, \quad (2)$$

where  $a$  collects the Sr systematic corrections (density, lattice, second-order Zeeman and BBR shifts). The Yb clock’s absolute frequency  $f_{\text{Yb}}$  (578 nm) relates to the 1156 nm comb branch by

$$f_{\text{comb}}^{1156} = \frac{1}{2} (1 + b) f_{\text{Yb}}. \quad (3)$$

The beat is converted to fractional units with the transfer frequency  $F_{1550} = 193.3992$  THz. The static gravitational-redshift correction enters the ratio inversion as a multiplicative factor,

$$(\text{Yb/Sr})_{\text{corr}} = (\text{Yb/Sr})_{\text{raw}} \times (1 + \delta_g), \quad \delta_g = -3.116 \times 10^{-15}, \quad (4)$$

derived from the leveled geopotential difference  $\Delta W = 280.042 \pm 0.085 \text{ m}^2 \text{ s}^{-2}$  (Methods).

## B Data processing and segment definition

The counters sample the beat every second in phase-averaging mode. Samples deviating by more than 10 Hz from the running median are treated as jump points and removed; the retained data are further trimmed at both ends by an endpoint screen that drops points beyond 1% of the peak-to-peak range of the record. The data set analyzed here consists of the longest jump-free record of each accepted segment after this screening. Table 2 lists the per-segment retained counts, ratio deviations and statistical uncertainties; the category column marks the five segments whose stability fits are model-limited.

## C Statistical combination models

**Fixed-effect WLS.** Under the assumption that all segments estimate one common ratio with uncertainties  $u_i$ ,

$$\mu_{\text{WLS}} = \frac{\sum_i w_i R_i}{\sum_i w_i}, \quad w_i = 1/u_i^2, \quad u_{\text{WLS}} = \left( \sum_i w_i \right)^{-1/2},$$

with  $\chi^2 = \sum_i (R_i - \mu_{\text{WLS}})^2 / u_i^2$  and  $\chi_{\text{red}}^2 = \chi^2 / (N - 1)$ . For the analyzed data set,  $\chi_{\text{red}}^2 = 4.017$  ( $N = 16$ , 15 degrees of freedom,  $p = 2.3 \times 10^{-7}$ ), so the fixed-effect uncertainty under-describes the observed scatter.

Table 2: Per-segment data of the 16 segments analyzed in this work. Ratio deviations  $y_i = R_i/R_1 - 1$  are relative to the first segment;  $u(y_i)$  is the statistical uncertainty of the segment ratio; “Fit” is *est.* for segments whose stability fit lies in the white-frequency-noise band and *lim.* for model-limited fits.

| Group | $n_{\text{valid}}$ | $y_i$ ( $10^{-18}$ ) | $u(y_i)$ ( $10^{-18}$ ) | Fit  |
|-------|--------------------|----------------------|-------------------------|------|
| 1     | 68 009             | +0.00                | 1.113                   | est. |
| 2     | 29 481             | -3.36                | 1.178                   | lim. |
| 3     | 42 891             | +2.00                | 1.984                   | est. |
| 4     | 64 280             | -4.61                | 0.951                   | est. |
| 5     | 19 532             | -3.38                | 6.040                   | est. |
| 6     | 52 140             | -1.15                | 0.513                   | lim. |
| 7     | 39 526             | +5.31                | 1.173                   | lim. |
| 8     | 43 565             | +0.97                | 4.509                   | est. |
| 10    | 57 938             | -0.76                | 1.230                   | est. |
| 11    | 35 824             | +2.61                | 2.301                   | est. |
| 12    | 40 346             | +0.73                | 1.715                   | lim. |
| 13    | 152 994            | -3.09                | 1.694                   | est. |
| 14    | 68 618             | +1.97                | 1.377                   | est. |
| 15    | 57 649             | -0.25                | 1.293                   | est. |
| 16    | 147 640            | -0.77                | 0.607                   | lim. |
| 17    | 65 983             | -0.89                | 2.142                   | est. |

**Birge scaling.** The Birge ratio  $B = \sqrt{\chi_{\text{red}}^2} = 2.004$  scales all segment uncertainties by a common factor,  $u_i \rightarrow Bu_i$ ; it assumes the relative sizes of the  $u_i$  are correct but their overall scale is not.

**Mandel–Paule.** An excess scatter  $\xi$  common to all segments is added in quadrature,  $u_{i,\text{eff}}^2 = u_i^2 + \xi^2$ , and  $\xi$  is solved from  $\chi_{\text{red}}^2(\xi) = 1$ . For the analyzed data set,  $\xi = 2.50 \times 10^{-18}$ .

**Bayesian random effect.** The same observation model,  $R_i \sim \mathcal{N}(\mu, u_i^2 + \xi^2)$ , is treated hierarchically:  $\mu$  and  $\xi$  are both inferred, with flat priors on  $\mu$  and on  $\log \xi$ , and the joint posterior  $p(\mu, \xi \mid \{R_i, u_i\})$  is marginalised numerically over a 40-decade  $\log\text{-}\xi$  grid. The reported central value is the posterior mean of  $\mu$  and the statistical uncertainty is its posterior standard deviation; for the analyzed data set,  $\xi = 2.83 \times 10^{-18}$ . The excess scatter quantified here also absorbs the correlated component of the time-varying gravitational-redshift term (Methods), which is carried separately as an uncertainty item rather than corrected.

## D Leveling reduction

The orthometric height difference between benchmarks follows from the leveled normal heights after the normal-to-orthometric conversion, with gravity observed at every leveling point. The geopotential difference is integrated segment by segment,  $\Delta W_{AB} = -\sum_i g_i \Delta H_i$ , and combined with the two local offsets between benchmark and clock platform (obtained by trigonometric leveling) and the platform-to-transition offsets from the clock design parameters. The resulting atomic-transition geopotential difference is  $\Delta W = 280.042 \pm 0.085 \text{ m}^2 \text{ s}^{-2}$ , equivalent to an optical fractional shift of  $-3.116 \times 10^{-15}$ .

## E Uncertainty budget details

The components of Table 1 are obtained as follows. The statistical term is the posterior standard deviation of the Bayesian random-effect model (Appendix C). The Sr and Yb terms are the published systematic uncertainties of the two clocks<sup>2,4</sup>; for the Yb clock the published evaluation is adopted. The static-potential term is  $u(\Delta W)/c^2$  from the leveling uncertainty  $0.085 \text{ m}^2 \text{ s}^{-2}$ , giving  $0.946 \times 10^{-18}$ . The fiber-link and comb terms are experiment-side upper bounds of  $1 \times 10^{-19}$  each, carried at  $0.1 \times 10^{-18}$  and  $0.1 \times 10^{-18}$  to preserve the experiment-side margin. The time-varying-redshift term is scaled from the spread of the segment-averaged gravitational shift across the campaign and is carried at  $0.5 \times 10^{-18}$ ; it is not applied as a correction. The total standard uncertainty,  $2.12 \times 10^{-18}$ , is the quadrature sum of the seven components.